

The computer chip

Computer chips are at the heart of all modern computers. They are found in phones, cars, and even washing machines. But what exactly are they and how are they made?

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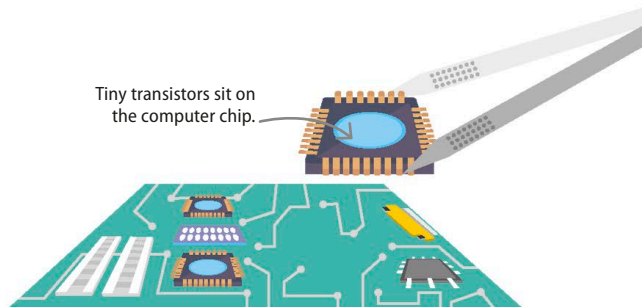
◀ 14–15 Computers are everywhere

Binary code 82–83 ▶

Logic gates 86–87 ▶

Integrated circuits

Computer chips are integrated circuits (ICs): silicon wafers with millions, or sometimes billions, of tiny components etched into them. They're much faster and smaller than circuits constructed from individual components, and cheaper to produce in large numbers. Chips are sealed into ceramic cases with metal pins connecting them to the rest of the computer. Integrated circuits can be made to carry out many tasks.



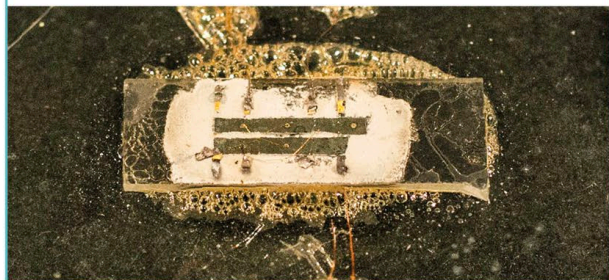
△ Transistor count

The building blocks of computer chips are transistors: tiny devices that are used to amplify or switch electric current. The higher the number of transistors, the more powerful a chip is.

REAL WORLD

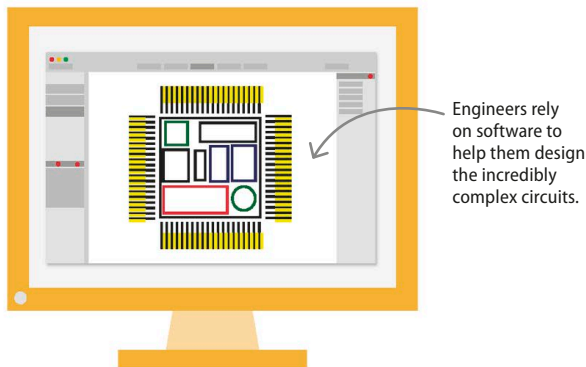
First integrated circuit

The first integrated circuit was created in 1958 by American electrical engineer Jack Kilby (1923–2005). Before Kilby's invention, machines used vacuum tubes, which were bulky and unreliable. Kilby's IC was based on tiny transistors, and all the parts were made on the one piece of material: the integrated chip was born.



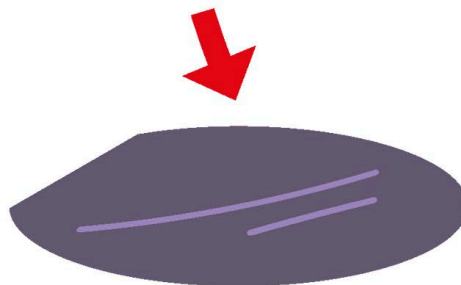
Manufacturing chips

Chips are manufactured in semiconductor wafer fabrication plants, known as fabs. Each fab has a "clean room" containing air with almost all dust particles filtered out, as even one dust mote can ruin a chip. Workers in the clean room wear suits and masks to protect the chips from the hair, skin cells, and any other potential contaminants that humans shed.



△ 1. Design

Microprocessor circuits are designed by teams of engineers who use software to define how the circuit should behave. The software then translates this definition into a layout of components.



△ 2. Wafers

The circuits are created from sheets of pure silicon. Dozens of chips are created at the same time from a single circular wafer.